NASPIER IMU BOARD V2

SENSOR_BOARD

File: SENSOR_IMU.kicad_sch

NASPIER IMU FFC

FLEX_CABLE

File: FLEX_CABLE_sch.kicad_sch

NASPIER CORE

CORE

File: CORE.kicad_sch

NASPIER CARRIER

CARRIER

File: CARRIER.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**SYSTEM-EXPOSE**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 1 Total 16
		REV0 VER0	A3	VAP-SAN

HIROSE 34P
FFC CONNECTOR

CORE_H743

STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)

File: CORE_H743.kicad_sch

CORE_POWER

IMU BUSSES 3.3V
REGULATORS

FC & AUX 3.3V
REGULATORS

File: CORE_POWER.kicad_sch

CORE_ONBOARD

AT24C64D
EEPROM-I2C

FRAM
SPI

ICM-45686
IMU-SPI

MS5611
BARO-I2C

File: CORE_ONBOARD.kicad_sch

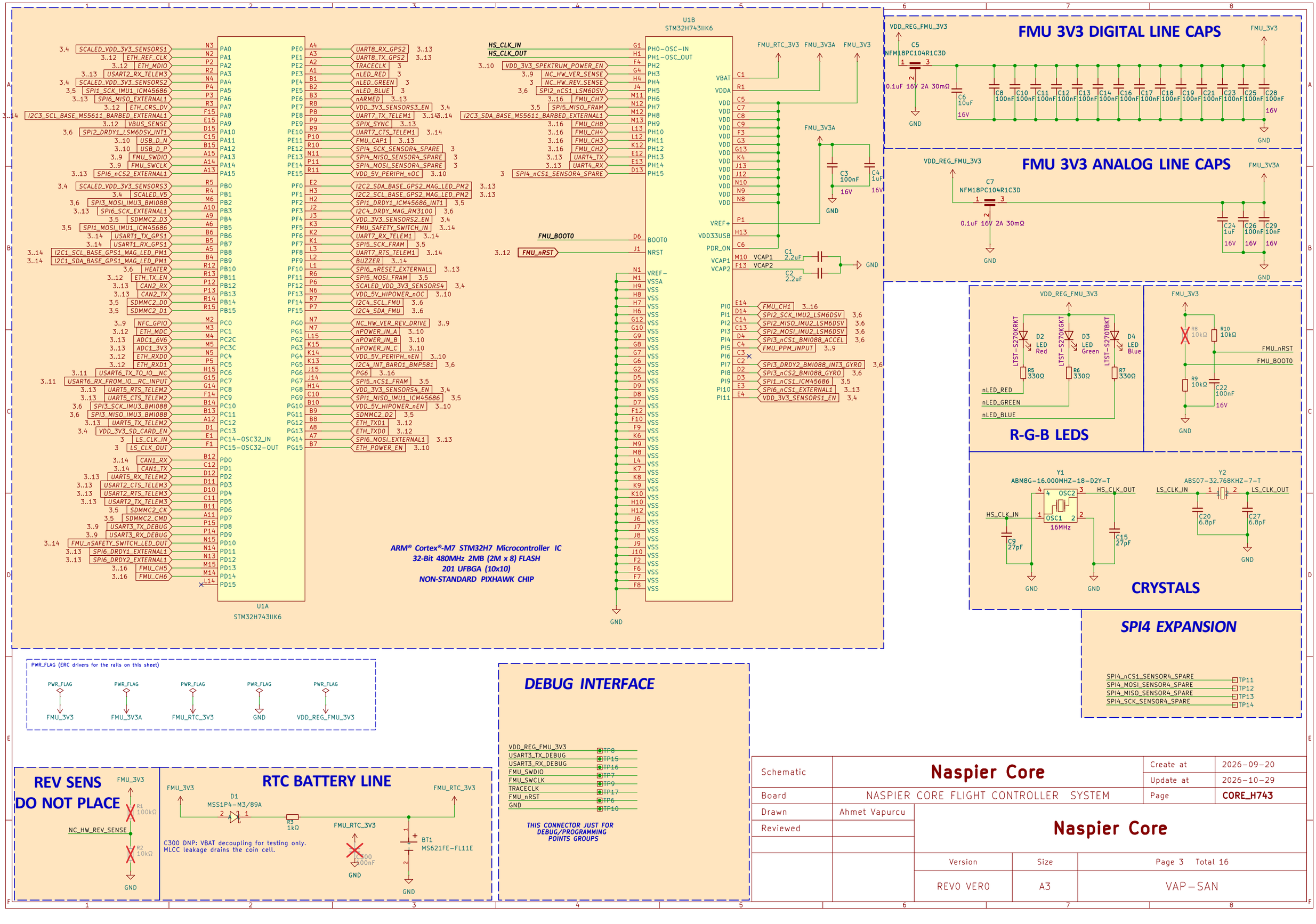
CORE_CONNECTORS

HIROSE 100P
CONNECTOR

HIROSE 50P
CONNECTOR

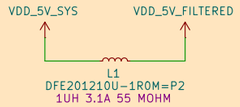
File: CORE_CONNECTORS.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	CORE BOARD
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size / Page 2 Total 16
		REV0 VER0	A4 / VAP-SAN

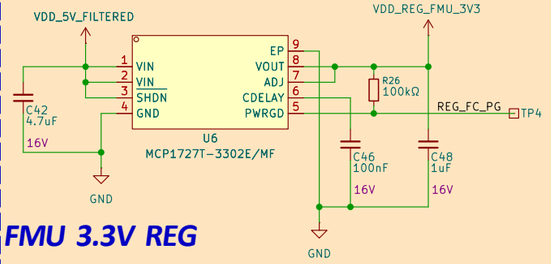


Schematic	Naspier Core	Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Update at	2026-10-29
Drawn	Ahmet Vapurcu	Page	CORE_H743
Reviewed			
	Naspier Core		
	Version: REV0 VER0	Size: A3	Page 3 Total 16
			VAP-SAN

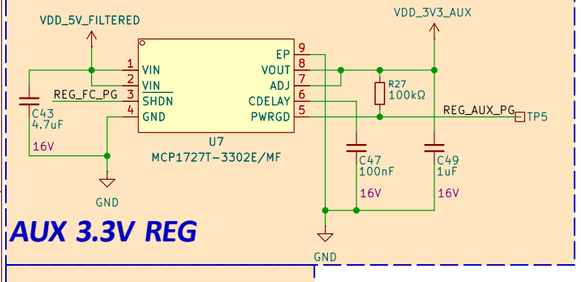
5V LINE FILTER



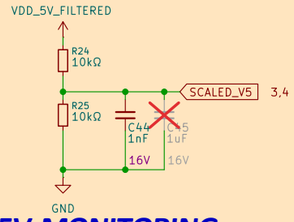
FMU 3.3V REG



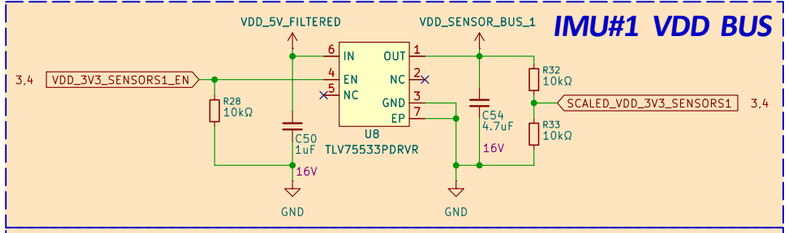
AUX 3.3V REG



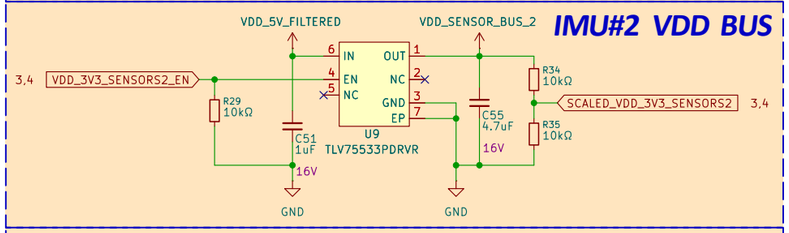
5V MONITORING



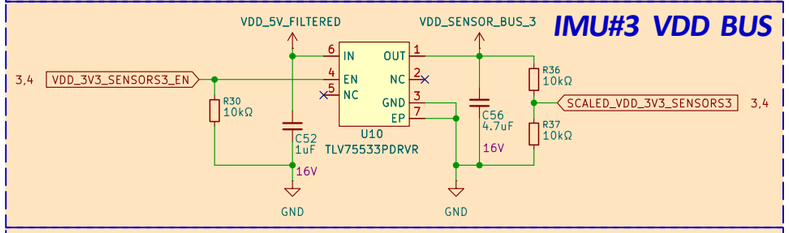
IMU#1 VDD BUS



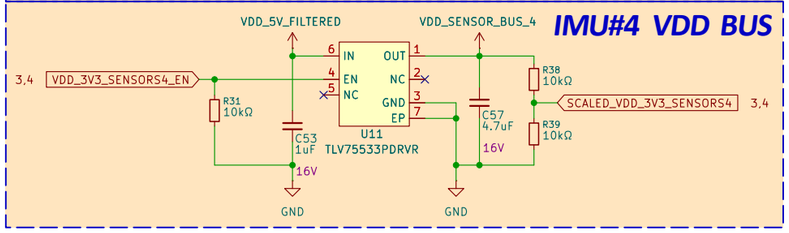
IMU#2 VDD BUS



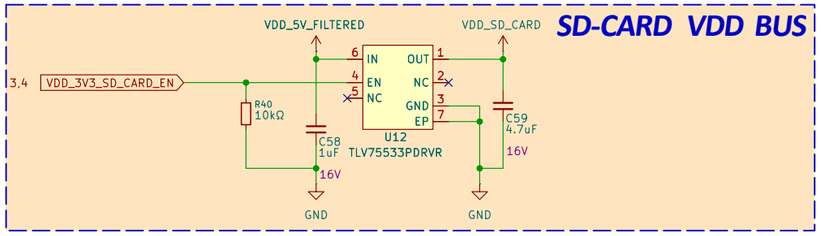
IMU#3 VDD BUS



IMU#4 VDD BUS

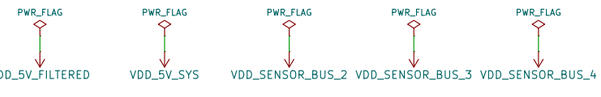


SD-CARD VDD BUS



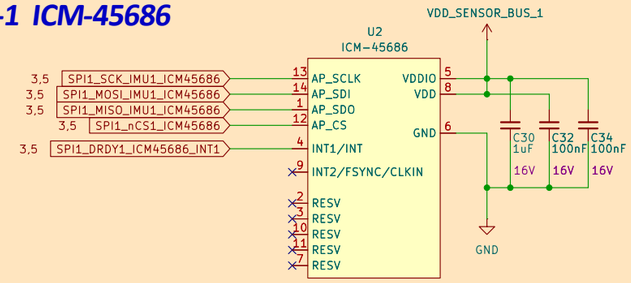
NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo
back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

PWR_FLAG (ERC drivers for the rails on this sheet)

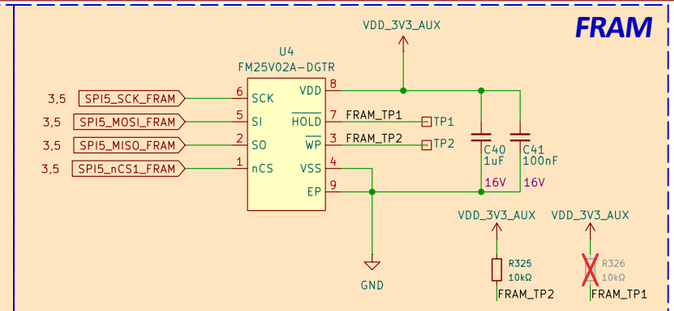


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 4 Total 16
		REV0 VER0	A3 / VAP-SAN

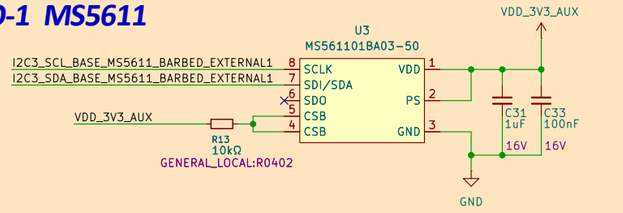
IMU-1 ICM-45686



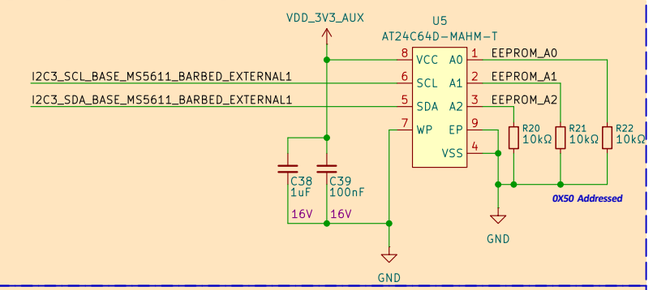
FRAM



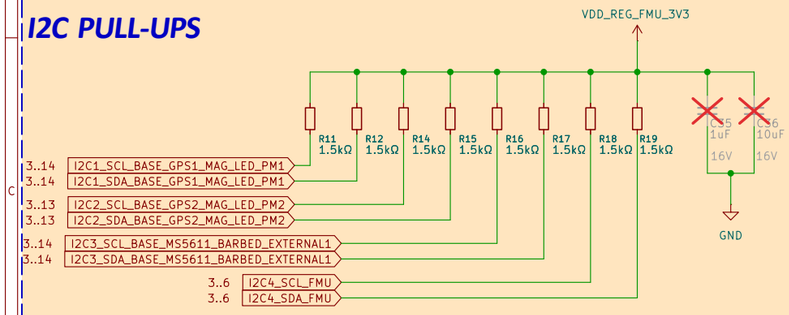
BARO-1 MS5611



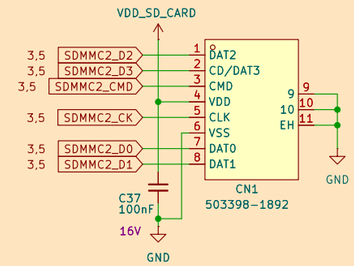
ONBOARD EEPROM



I2C PULL-UPS



SD-CARD



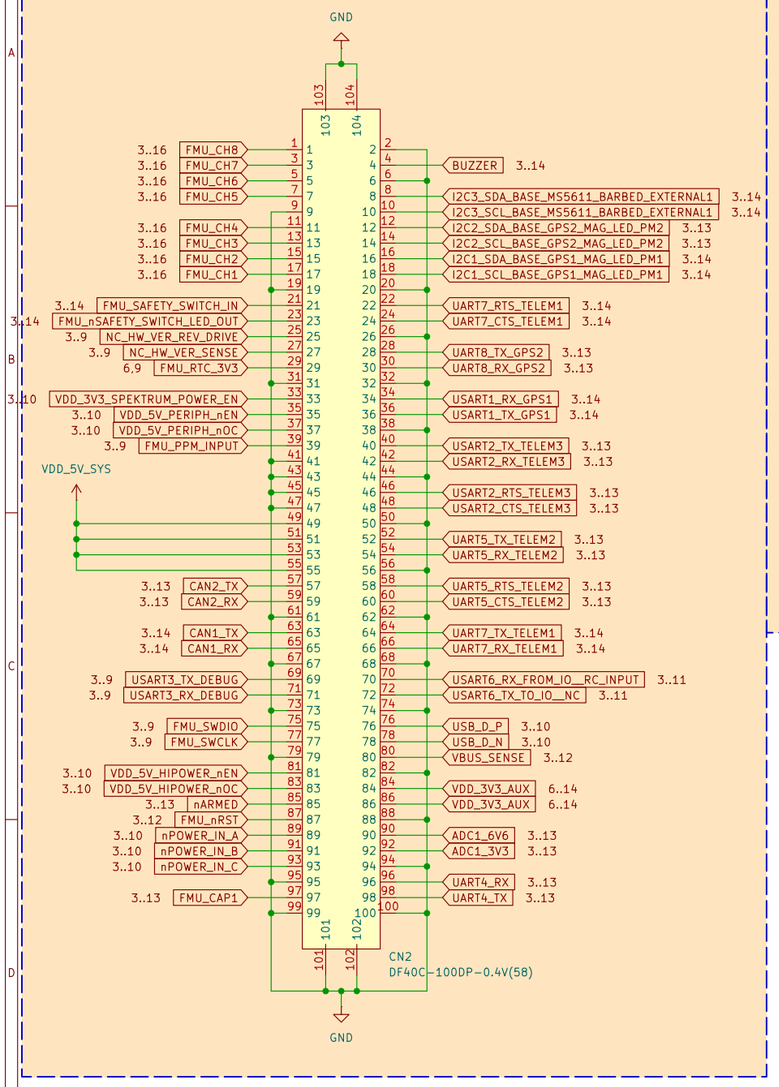
Push-In
Push-Out

PWR_FLAG (ERC drivers for the rails on this sheet)



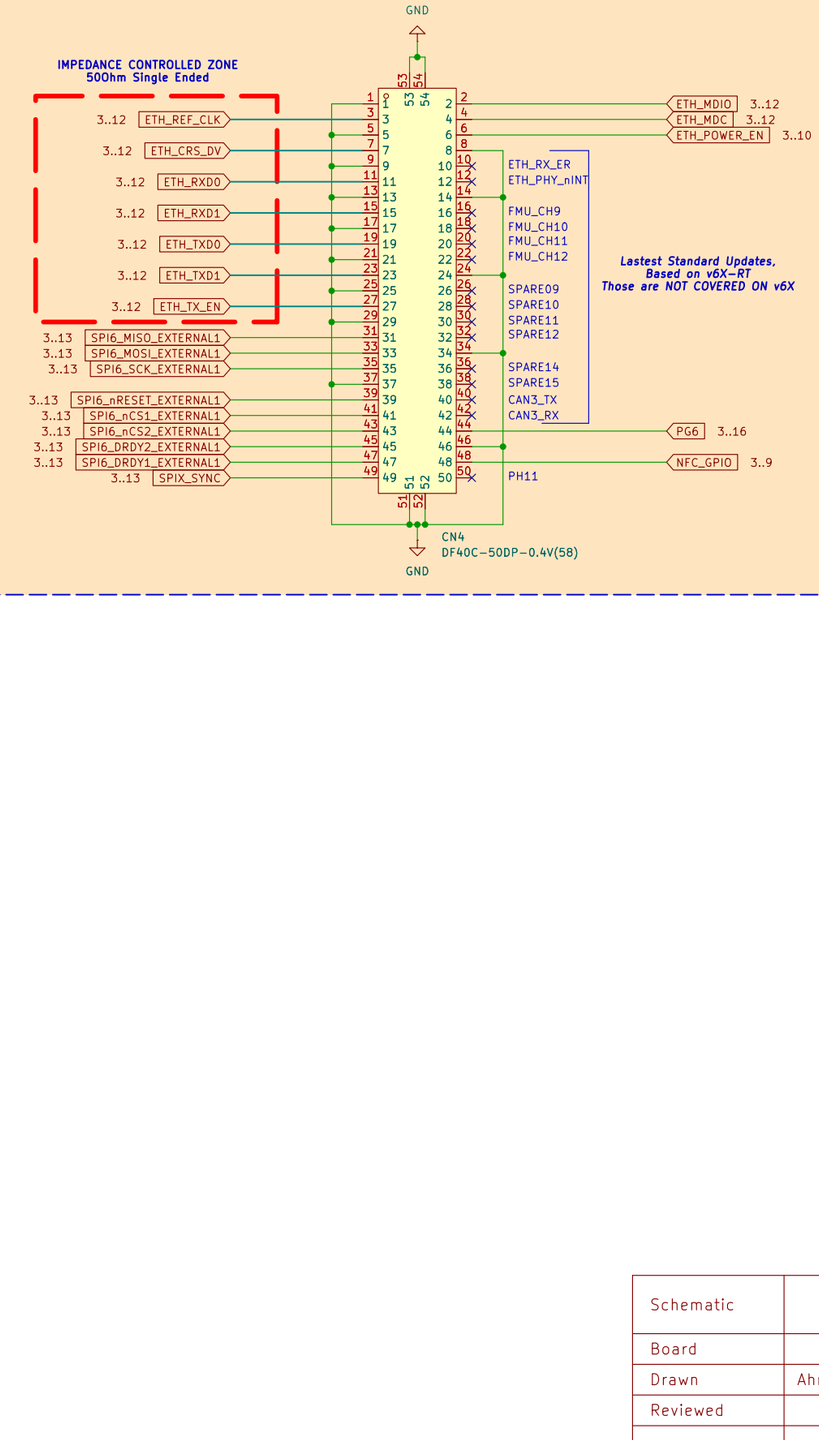
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_ELEMENTS**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 5 Total 16
		REVO VERO	A3 / VAP-SAN

PAB X1 100P HIROSE

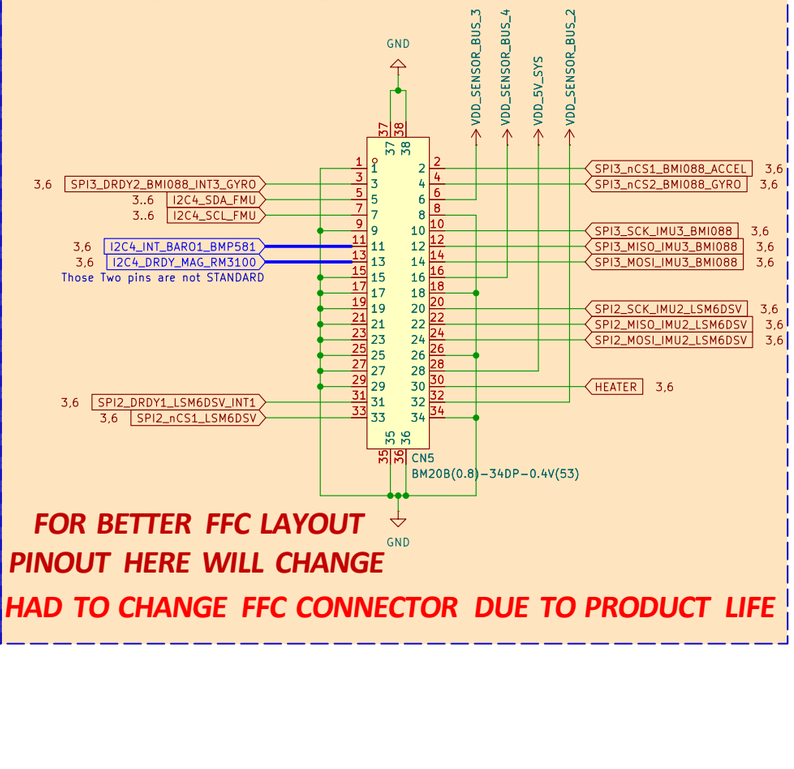


PAB X2 50P HIROSE

v6X-RT only (not on v6X cores): FMU_CH9-CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09-12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.



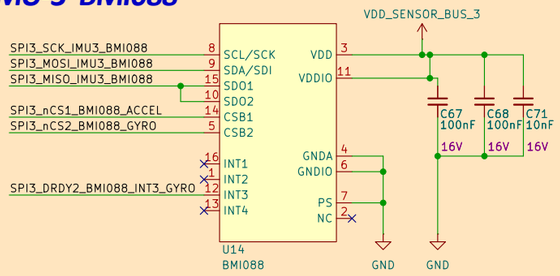
PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



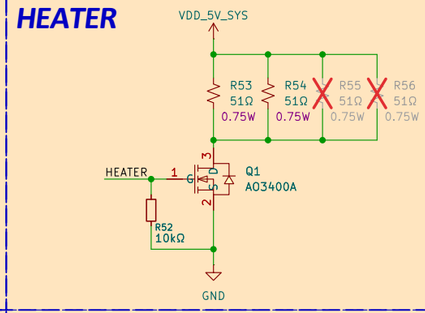
FOR BETTER FFC LAYOUT
PINOUT HERE WILL CHANGE
HAD TO CHANGE FFC CONNECTOR DUE TO PRODUCT LIFE

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 6 Total 16	
		REV0 VER0 / A3 / VAP-SAN	

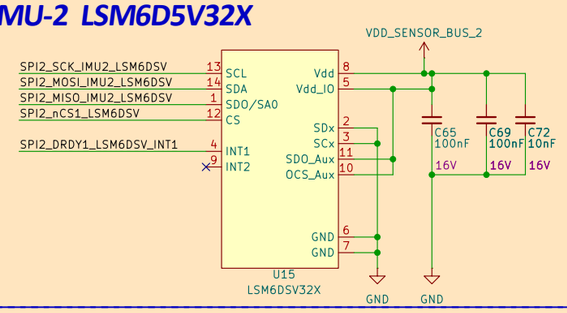
IMU-3 BMI088



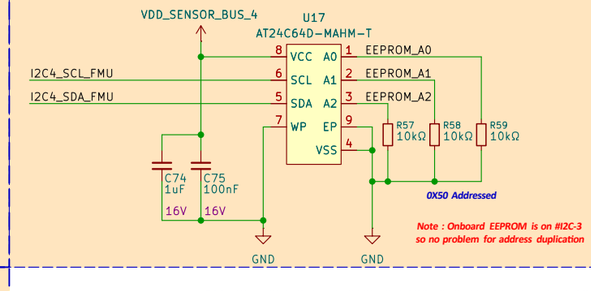
HEATER



IMU-2 LSM6D5V32X

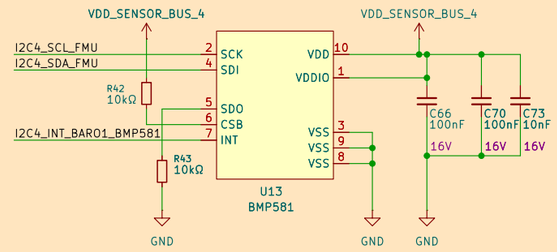


EEPROM

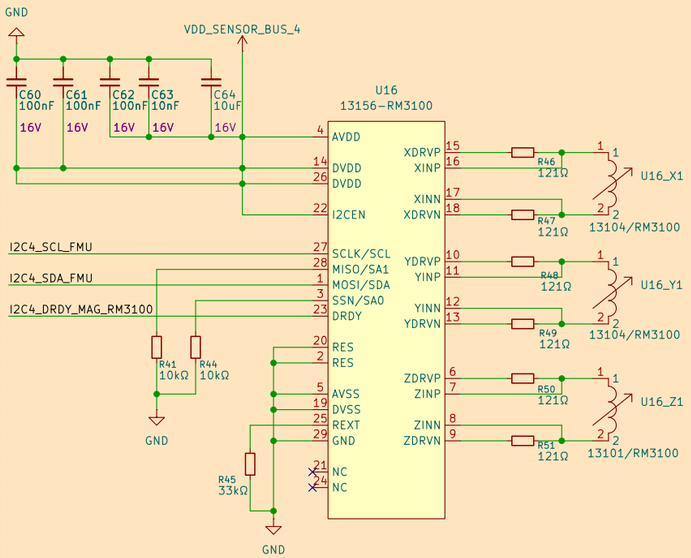


C11: R53–R56 Rohm LTR18EZPF51R0, 0.75 W wide-terminal 1206 (0612 footprint).
R55 / R56 DNP: fit them for more heater power when needed.

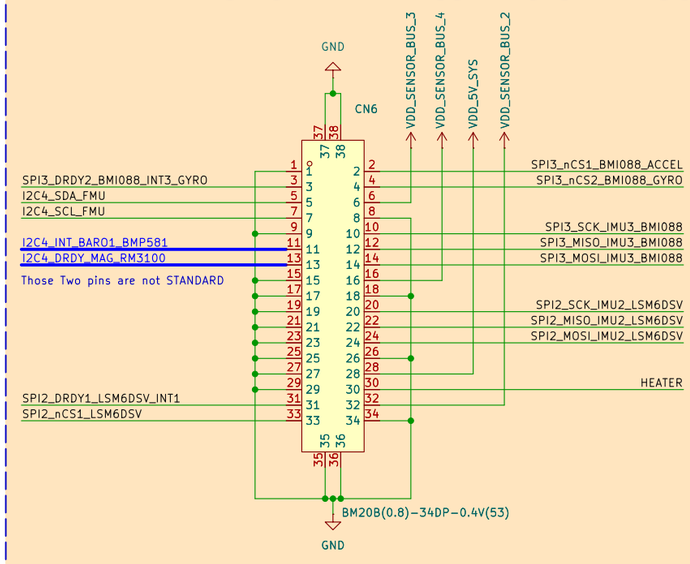
BARO-2 BMP581



MAG-RM3100



PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



FOR BETTER FFC LAYOUT
PINOUT HERE WILL CHANGE

Schematic	**Naspier Core**	Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Update at	2026-10-29
Drawn	Ahmet Vapurcu	Page	**SENSOR_BOARD**
Reviewed		**Naspier Core**	
		Version: REVO VERO	Size: A3
		Page 7 Total 16	VAP-SAN

CARRIER_B2B

File: CARRIER_B2B.kicad_sch

CARRIER_CONN_1

File: CARRIER_CONN_1.kicad_sch

CARRIER_IO_MCU

File: CARRIER_IO_MCU.kicad_sch

CARRIER_POWER

File: CARRIER_POWER.kicad_sch

CARRIER_CONN_2

File: CARRIER_CONN_2.kicad_sch

CARRIER_ELEMENTS

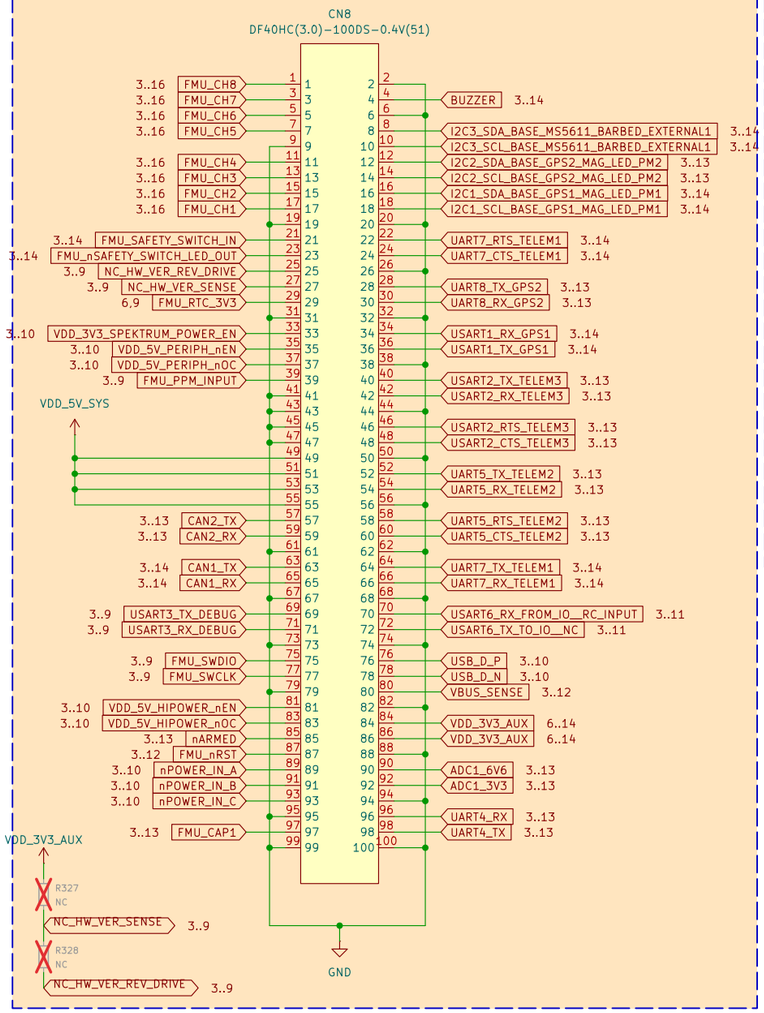
File: CARRIER_ELEMENTS.kicad_sch

CARRIER_PWM

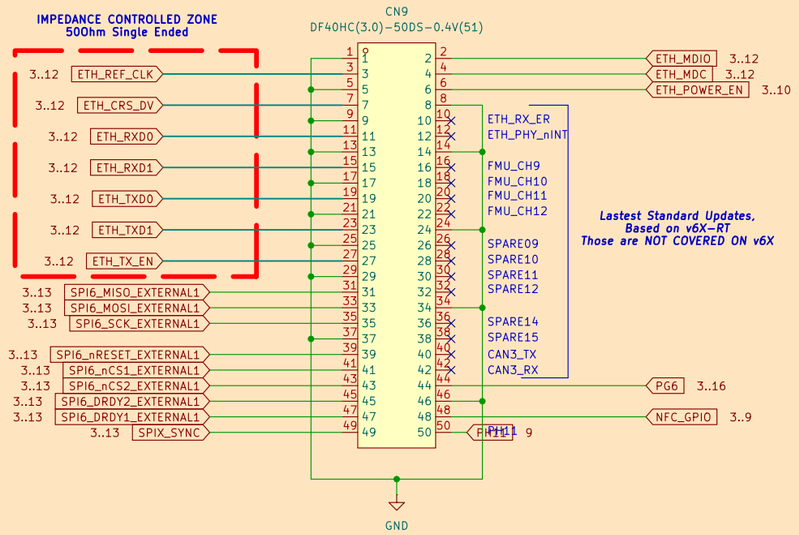
File: CARRIER_PWM.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER BOARD**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 8 Total 16
		REV0 VER0	A4	VAP-SAN

PAB X1 100P HIROSE

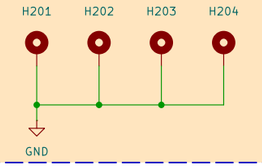


PAB X2 50P HIROSE



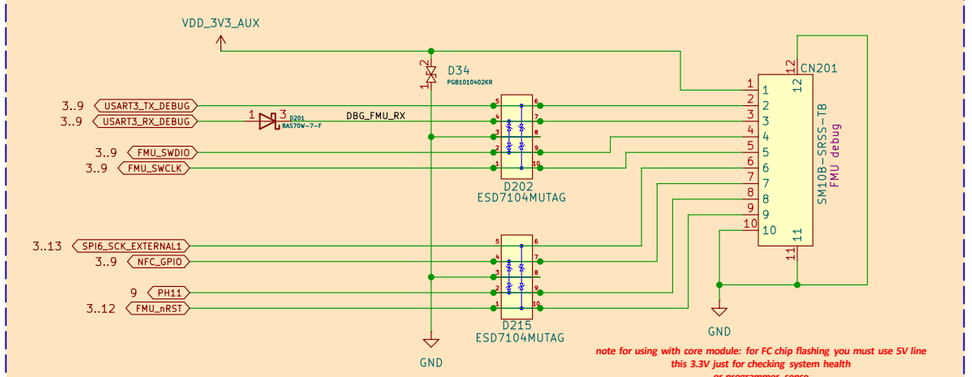
v6X-RT only (not on v6X cores): FMU_CH9–CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09–12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.

MountingHole M2 x4
CORE SCREWED STANDOFFS

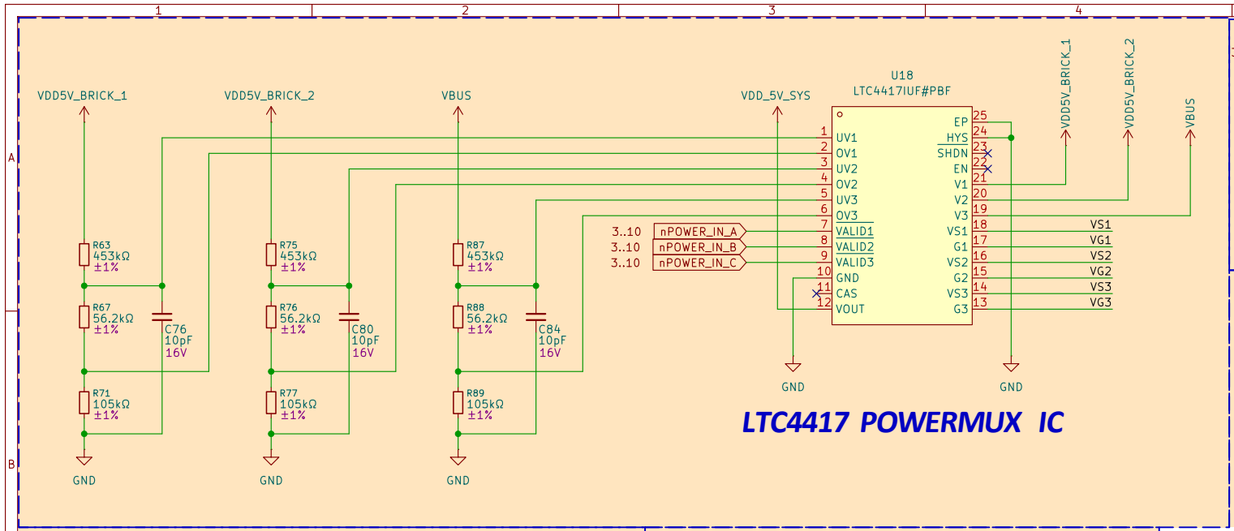


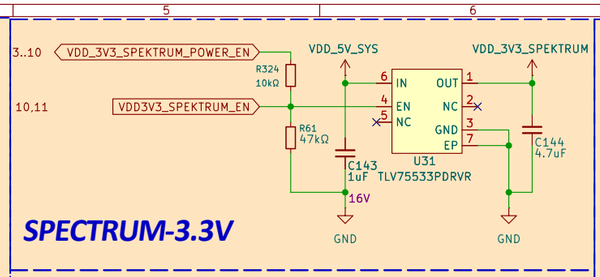
HW_VER_SENSE divider (RC05 page 2, R13 / R18): R327 from FMU_VDD_3V3 (X1 84 / 86 = VDD_3V3_AUX), R328 to HW_VER_REV_DRIVE. Both DNP; values set the baseboard version. Net names keep the NC_ prefix.

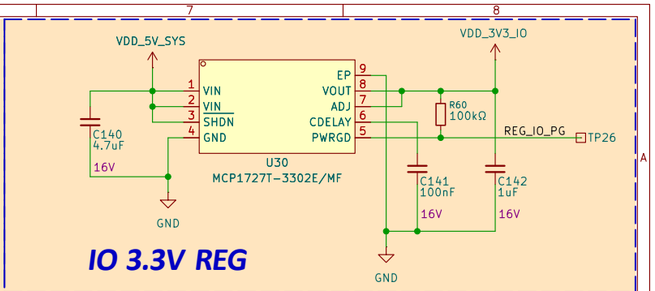
FMU DEBUG

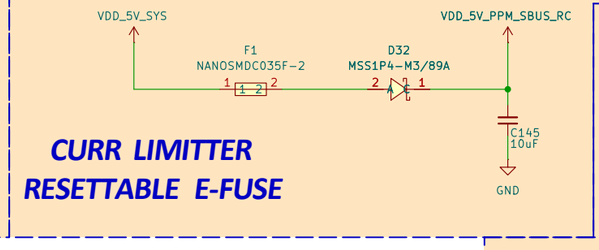


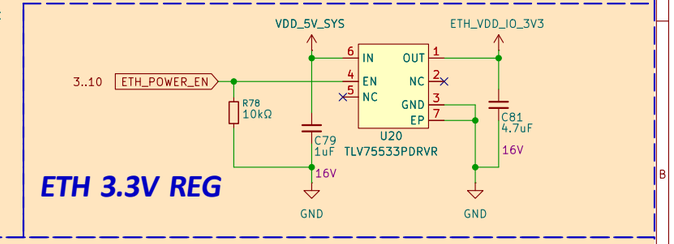
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 9 Total 16
		REVO VERO	A3 / VAP-SAN

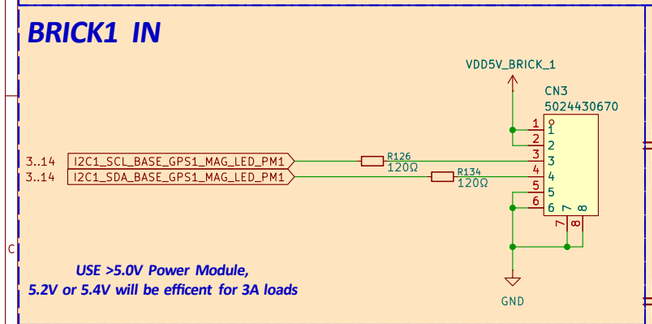


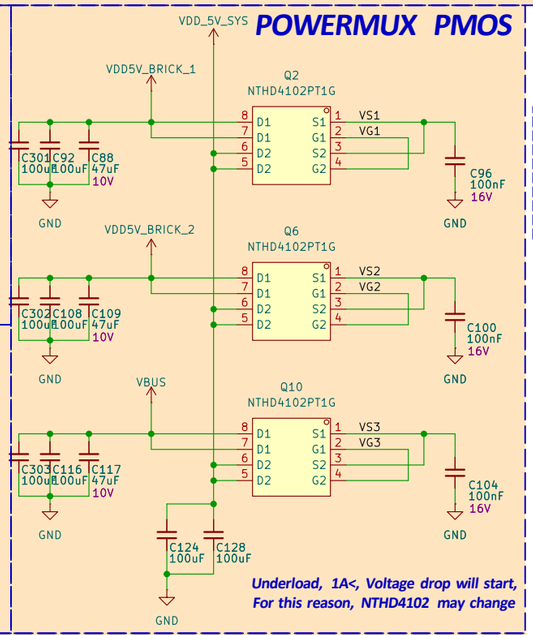


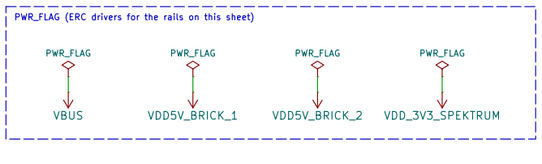


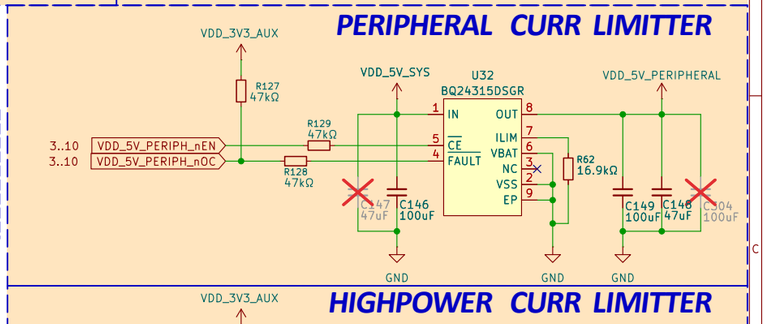


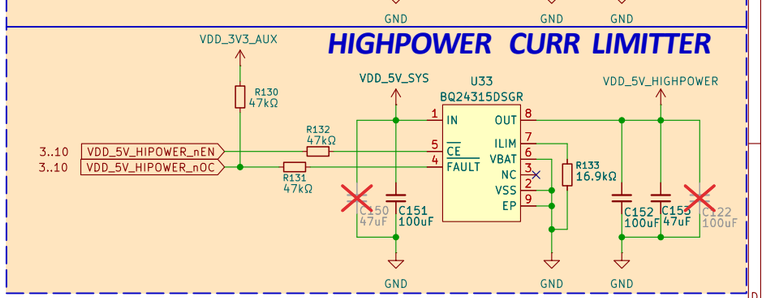


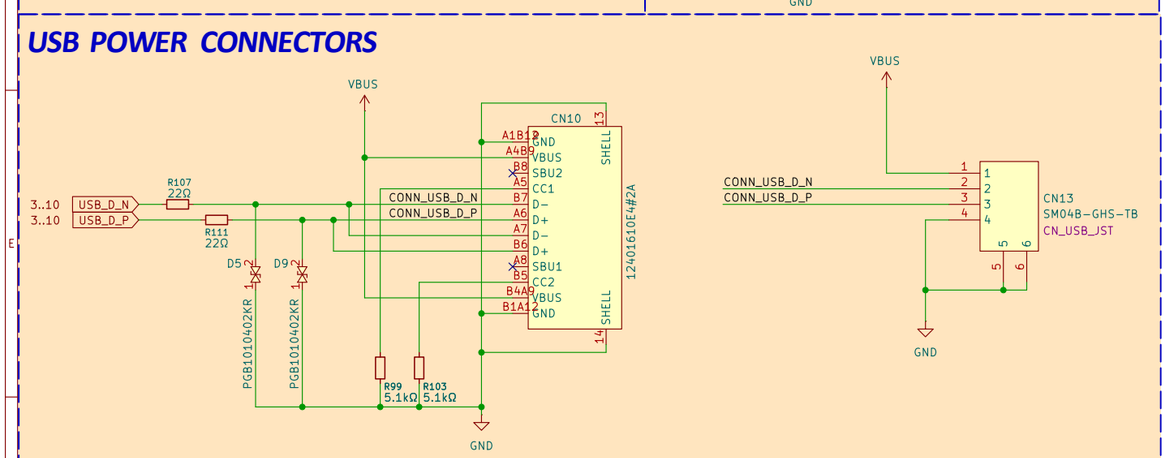










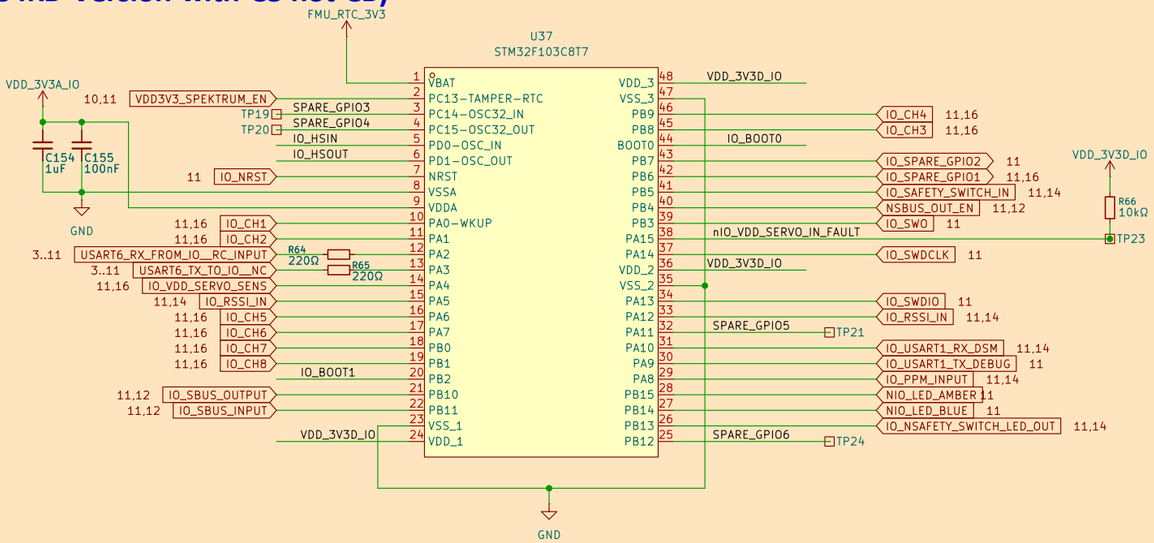


NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

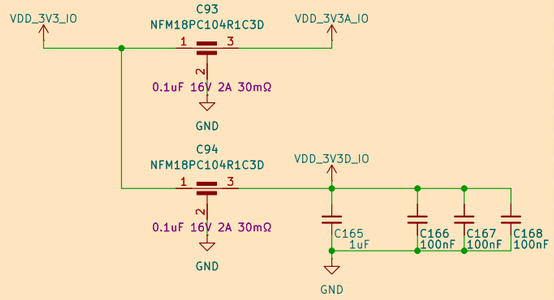
L6 (RC05 page 8): U31 EN = VDD3V3_SPEKTRUM_EN, driven by the IO MCU (PC13) directly and by the FMU (PH2, VDD_3V3_SPEKTRUM_POWER_EN, DS-012) through R324 10 k; R61 47 k pull-down. Both are the regulator enable (the receiver has no chip enable): the IO MCU can power-cycle it for DSM bind.

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 10 Total 16
		REV0 VER0	A3	VAP-SAN

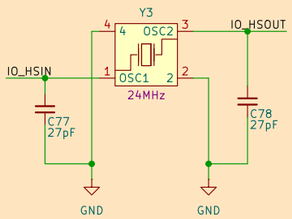
IO F103 CHIP
(64KB Version with C8 not CB)



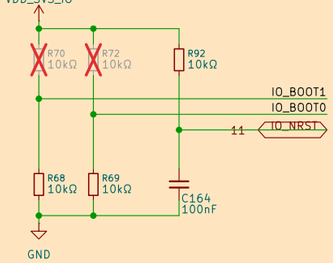
IO POWER FILTER

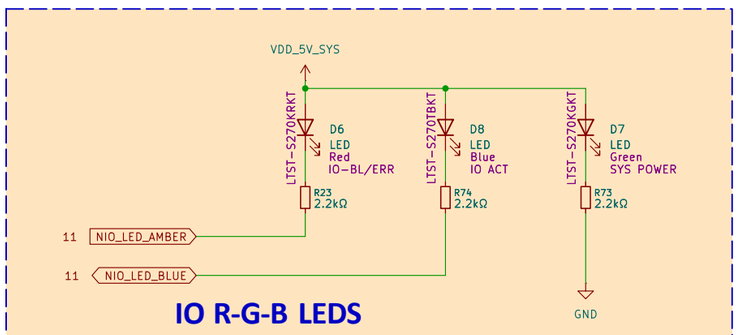


IO CRYSTAL



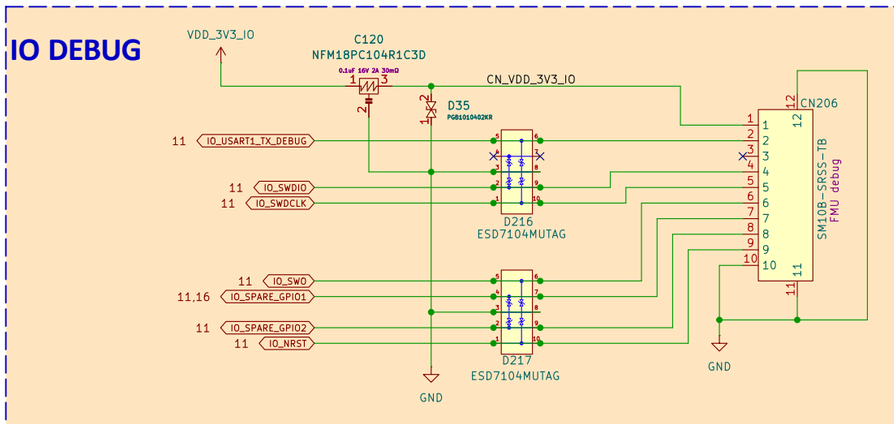
IO BOOT-NRST

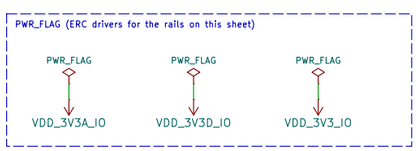




IO R-G-B LEDS

IO DEBUG





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_IO_MCU**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 11 Total 16
		REVO VERO	A3	VAP-SAN

CHECK (C17): RC05 drives ETH_NRESET from the MIC5332 POR output.
Here nRST has only R116 (10 k pull-up): the PHY relies on its internal POR.

v6X-RT only: ETH_RX_ER (X2 pin 10) and R112 are used with v6X-RT cores, not with v6X.

ETH_VDD_IO_3V3

C159 10uF, C158 100nF, C157 100nF

C160 100nF, C161 10nF

R114 1.5kΩ, R116 10kΩ

ETH_MDIO
ETH_NRST

C162 1uF, C163 470pF

ETH_VDDCR

U19
LAN8742AI-CZ

VDD2A
VDD1A
VDDIO
VDDCR
RXER/PHYAD0
CRS_DV/MODE2
MDIO
MDC
TXEN
nRST
TXD1
TXD0
RXD1/MODE1
RXD0/MODE0
nINT/REFCLKO
RXN
RXP
TXN
TXP
LED2/nINT/nPME/nINTSEL
LED1/nINT/nPME/REGOFF
XTAL2
XTAL1/CLKIN
RBIAS
EPAD

ETH_RX_ER
ETH_CRS_DV
ETH_MDIO
ETH_MDC
ETH_TX_EN
ETH_NRST
ETH_TXD1
ETH_TXD0
ETH_RXD1
ETH_RXD0
ETH_REF_CLK

R112 (NC), R113 22Ω, R117 22Ω, R118 22Ω, R120 22Ω

ETH_HSOUT
ETH_HSIN

R121 12.1kΩ

OSC1 25MHz Y4
C82 27pF
C83 27pF

ETH_VDD_IO_3V3

C85 100nF, R85 49.9Ω 1%, R86 49.9Ω 1%, R90 49.9Ω 1%, R91 49.9Ω 1%, C86 100nF

RX_N
RX_P
TX_N
TX_P

D27 ESD7104MUTAG

L2 DLW43MH201XK2L
L3 DLW43MH201XK2L

C87 100nF ISOLATION
C89 100nF ISOLATION
C90 100nF ISOLATION
C91 100nF ISOLATION

PHY_RX_N
PHY_RX_P
PHY_TX_N
PHY_TX_P

CN7
SM04B-GHS-TB
ETH

D11 LED Blue ETH Link/Act
LTST-S270TBKT
D10 LED
R84 10kΩ
R83 1kΩ
R82 1kΩ

ETH_VDD_IO_3V3
LTST-S270KRKT
D31 LED Red ETH 3V3 POWER
R154 1kΩ

MountingHole M3 x4 BASEBOARD HOLES

H205 H206 H207 H208

H205–H208: M3 baseboard holes, plain (no standoff):
3.1 mm drill, 5.0 mm plated pad to GND.

CARRIER EEPROM EXT SENSOR CAL.

VDD_3V3_AUX

U21
AT24C64D-MAHM-T

I2C3_SCL_BASE_MS5611_BARBED_EXTERNAL1
I2C3_SDA_BASE_MS5611_BARBED_EXTERNAL1

EEPROM_A0
EEPROM_A1
EEPROM_A2

R153 10kΩ, R151 10kΩ, R152 10kΩ

C114 1uF 16V, C115 100nF 16V

0X51 Addressed

VBUS SENS CIRCUIT

VBUS

R115 10kΩ, R119 10kΩ

TP18
FMU_BOOTLOADER

U22
MC74VHC1G32DFT1G

U26
MC74VHC1G09DFT1G

VDD_3V3_AUX
VDD_5V_SYS

R123 1kΩ

TP22
VBUS_SENSE

C132 100nF, C136 100nF

PWR_FLAG (ERC drivers for the rails on this sheet)
PWR_FLAG
ETH_VDD_IO_3V3

RC INVERT BUFFER

VDD_3V3_IO

C203 100nF

U201
74LVC2G240DP
INVENT BUFFER

RC_1A
RC_2Y
RC_OE2
NSBUS_OUT_EN
IO_SBUS_INPUT
IO_SBUS_OUTPUT

R203 220
R204 10k

NRST BUFFER

VDD_3V3_AUX

C208 100nF

nARMED
FMU_RST_REQ
FMU_nRST

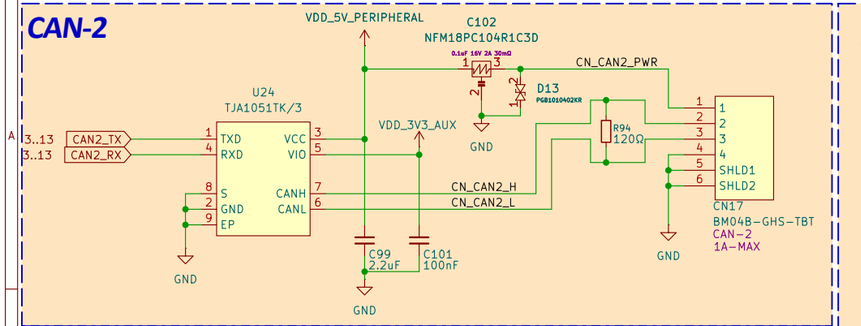
U202
74LVC1G38
(NAND, open drain)

Schematic	Naspier Core	Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Update at	2026-10-29
Drawn	Ahmet Vapurcu	Page	CARRIER_ELEMENTS
Reviewed			

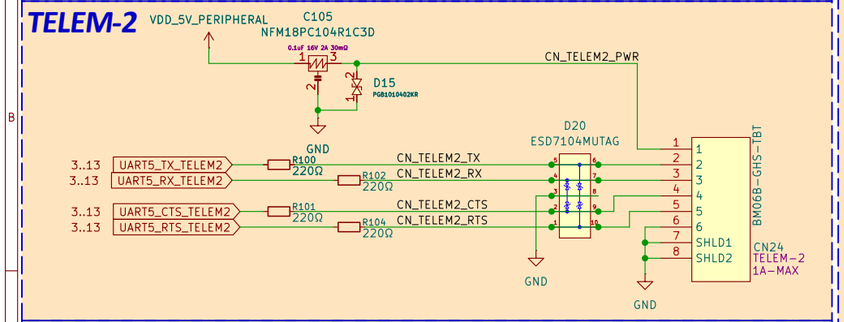
Naspier Core

Version	Size	Page 12 Total 16
REVO VERO	A3	VAP-SAN

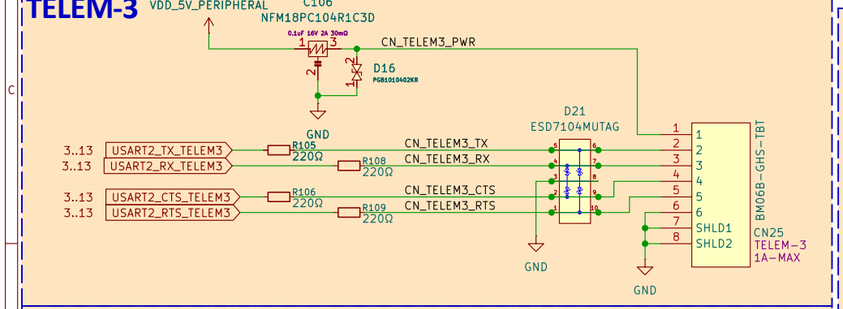
CAN-2



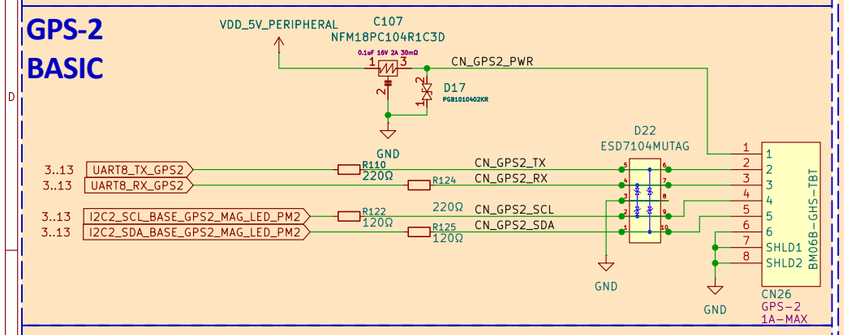
TELEM-2



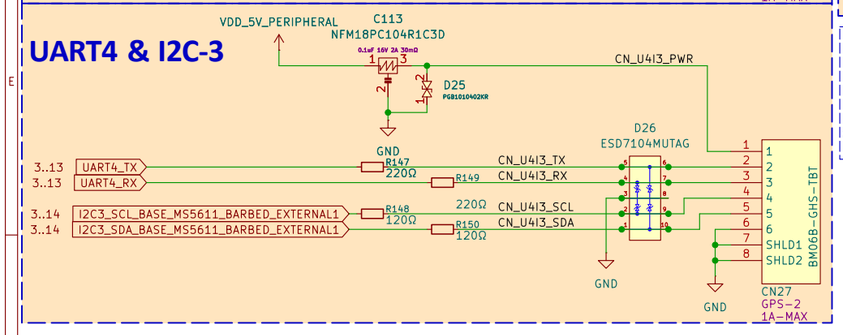
TELEM-3



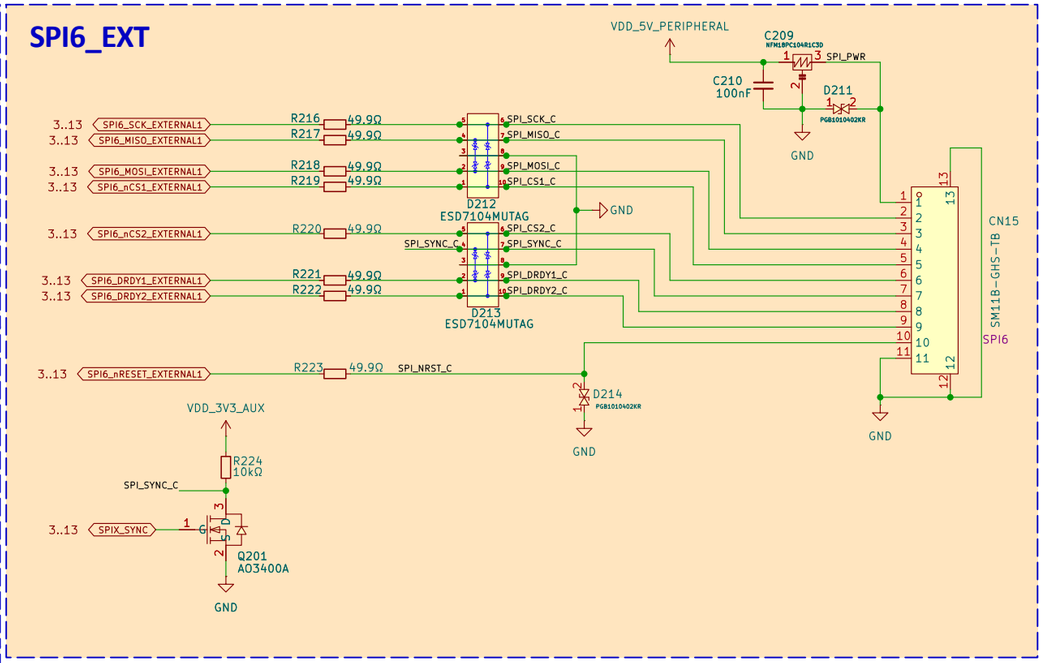
GPS-2 BASIC



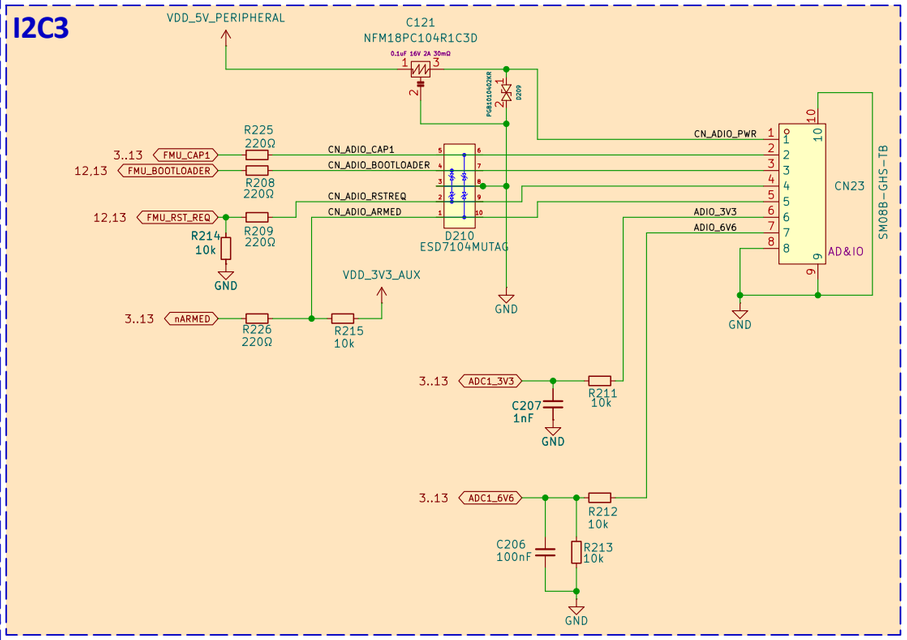
UART4 & I2C-3



SPI6_EXT



I2C3



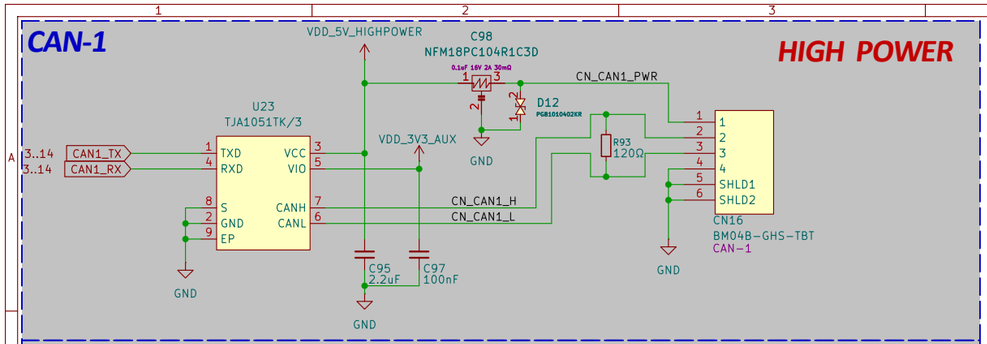
PWR_FLAG (ERC drivers for the rails on this sheet)



Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_CONN_1**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size
		REVO VERO	A3
		Page 13 Total 16	
		VAP-SAN	

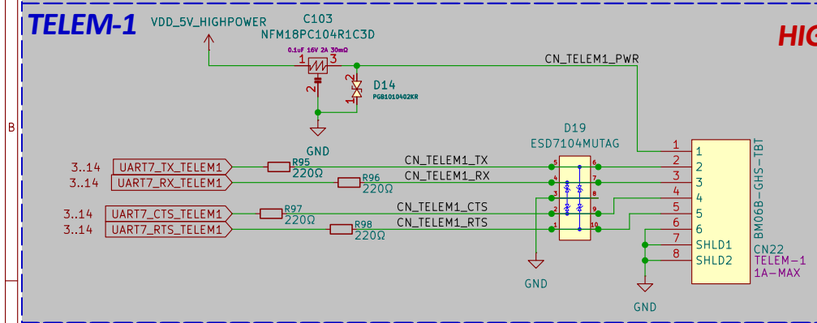
CAN-1

HIGH POWER



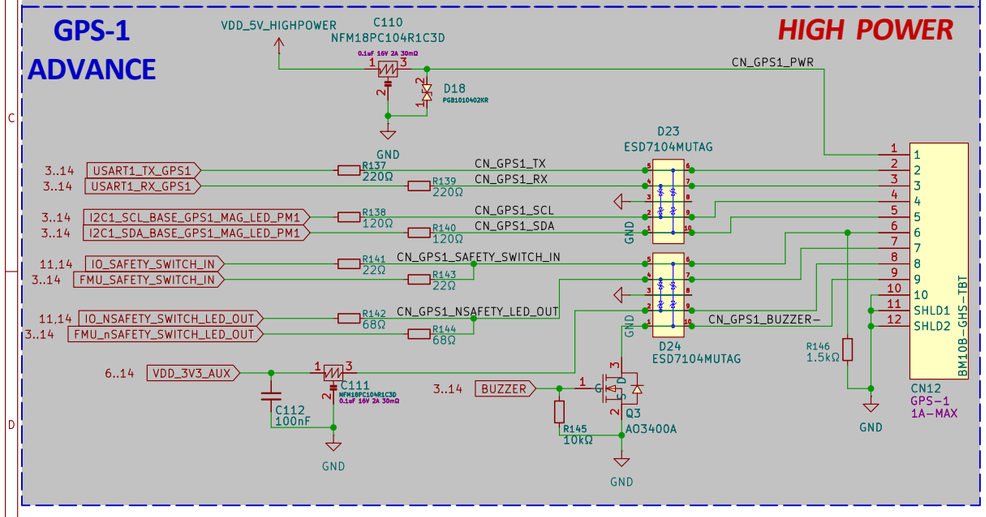
TELEM-1

HIGH POWER

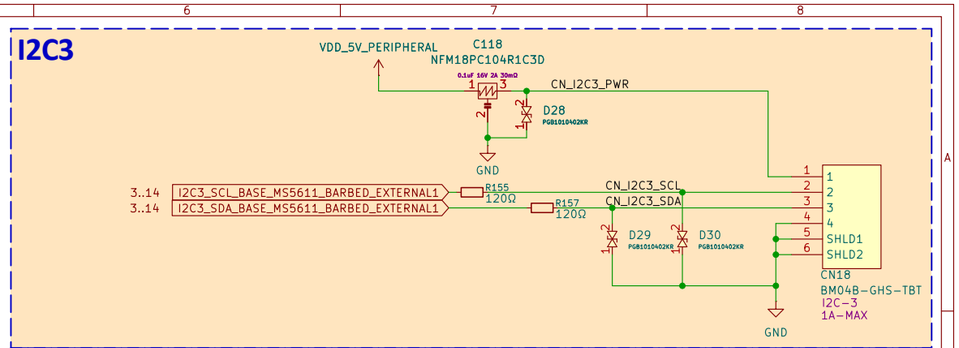


GPS-1 ADVANCE

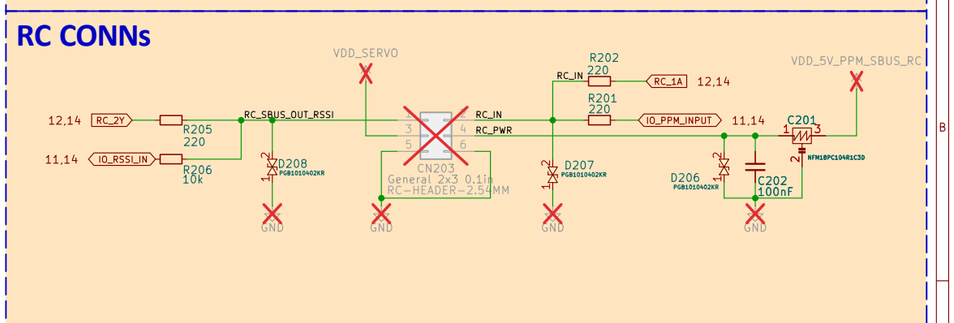
HIGH POWER



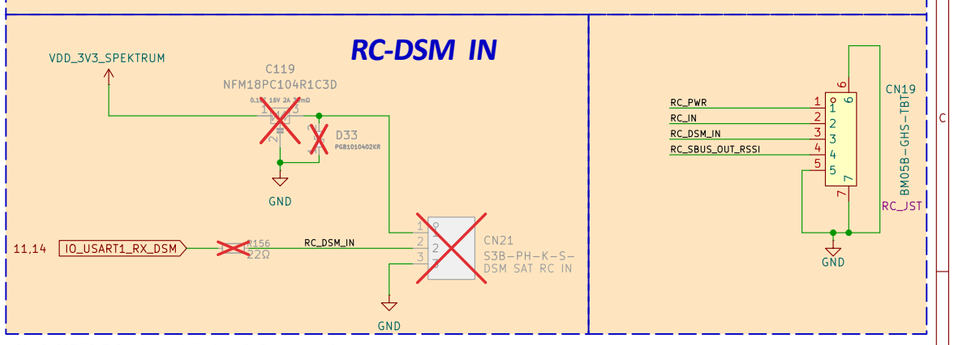
I2C3



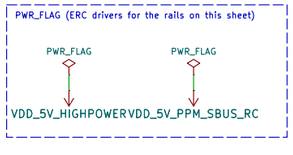
RC CONNs



RC-DSM IN

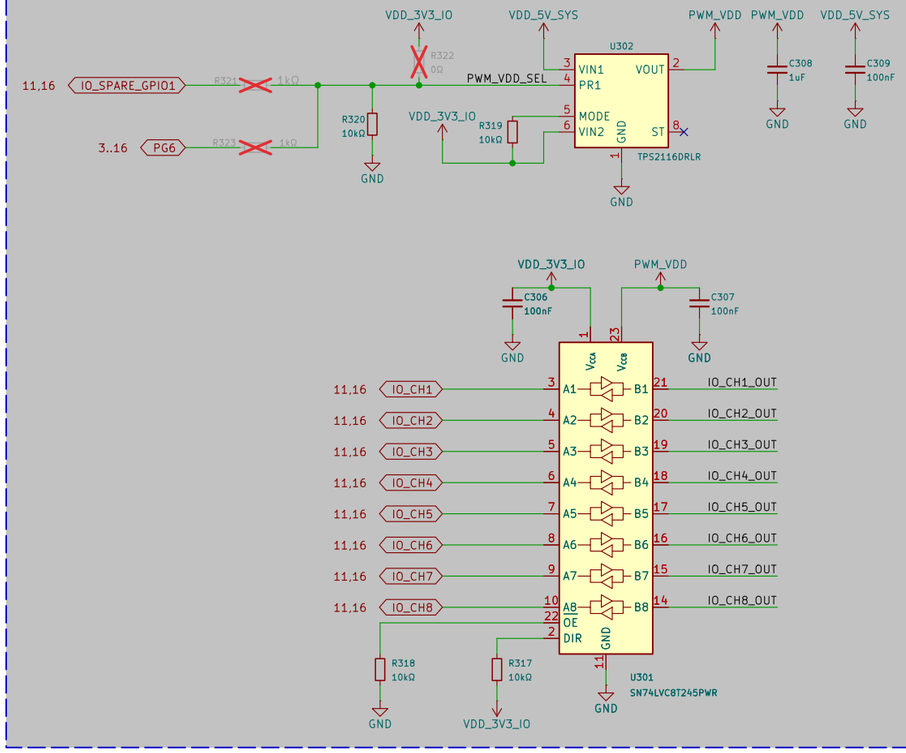


C4: CN203 (2x3 0.1 in header) stays DNP as an option.
R201 R202 R205 R206 C201 C202 D206–D208 are fitted: they also serve CN19 (GH RC IN).

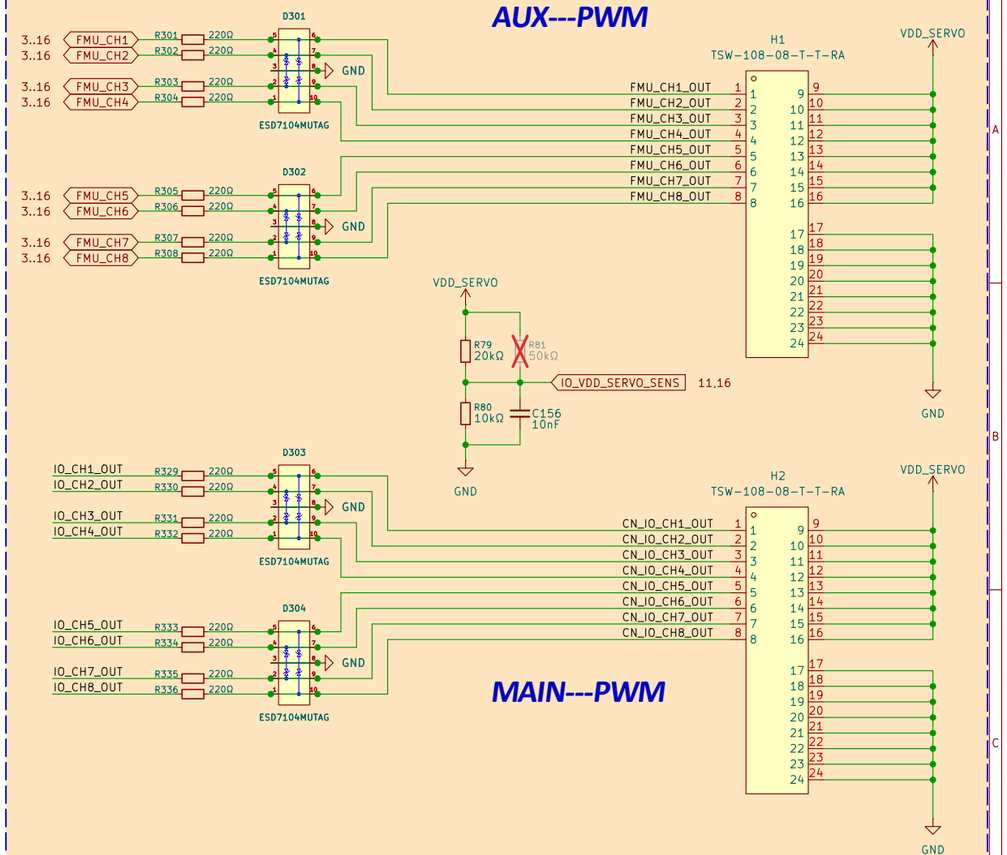


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_CONN_2**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 14 Total 16
		REV0 VER0	A3 / VAP-SAN

AUX PWM VOLTAGE LEVET OPT.



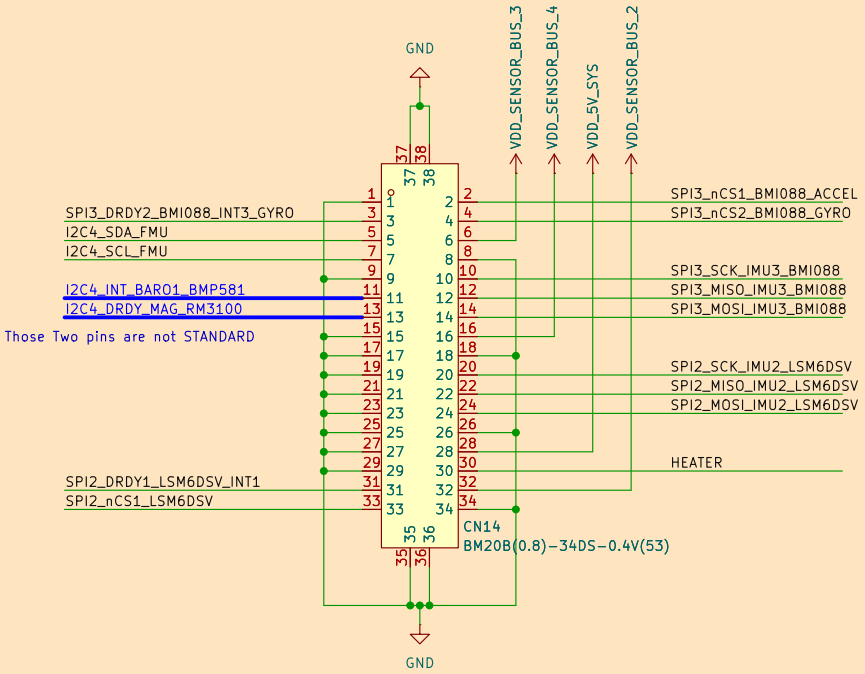
AUX---PWM

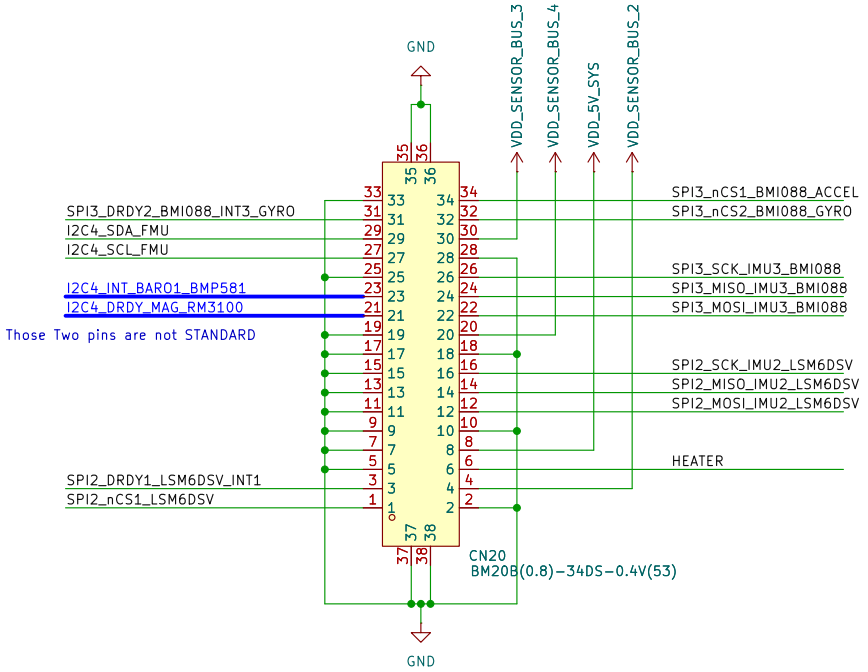


FMU Drive with PG6	DIGITAL SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
IO Drive with SPARE_GPIO1	DIGITAL SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
R322 FITTED	ANALOG SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
OTHERWISE	ANALOG SET	PWM_VDD_SEL : 0V	MAIN PWM : 3.3V

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_PWM**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 16 Total 16
		REV0 VER0	A3	VAP-SAN

PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**FLEX_CABLE**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 15 Total 16
		REV0 VER0	A4	VAP-SAN